HCI and Design

Today

Assignment 1 is graded

Assignment 3 is posted

Understanding prototype fidelity

What is Digital Prototyping?

Introduction to various digital prototyping tools

Reminder: What is a prototype?

A prototype is an incomplete, early version of a product

There are many approaches to building prototypes for software user interfaces

UI prototypes can be as simple as a drawing on a piece of paper or as complex as functional web application

- Or anywhere in between those extremes!

Lo-Fi vs. Hi-Fi

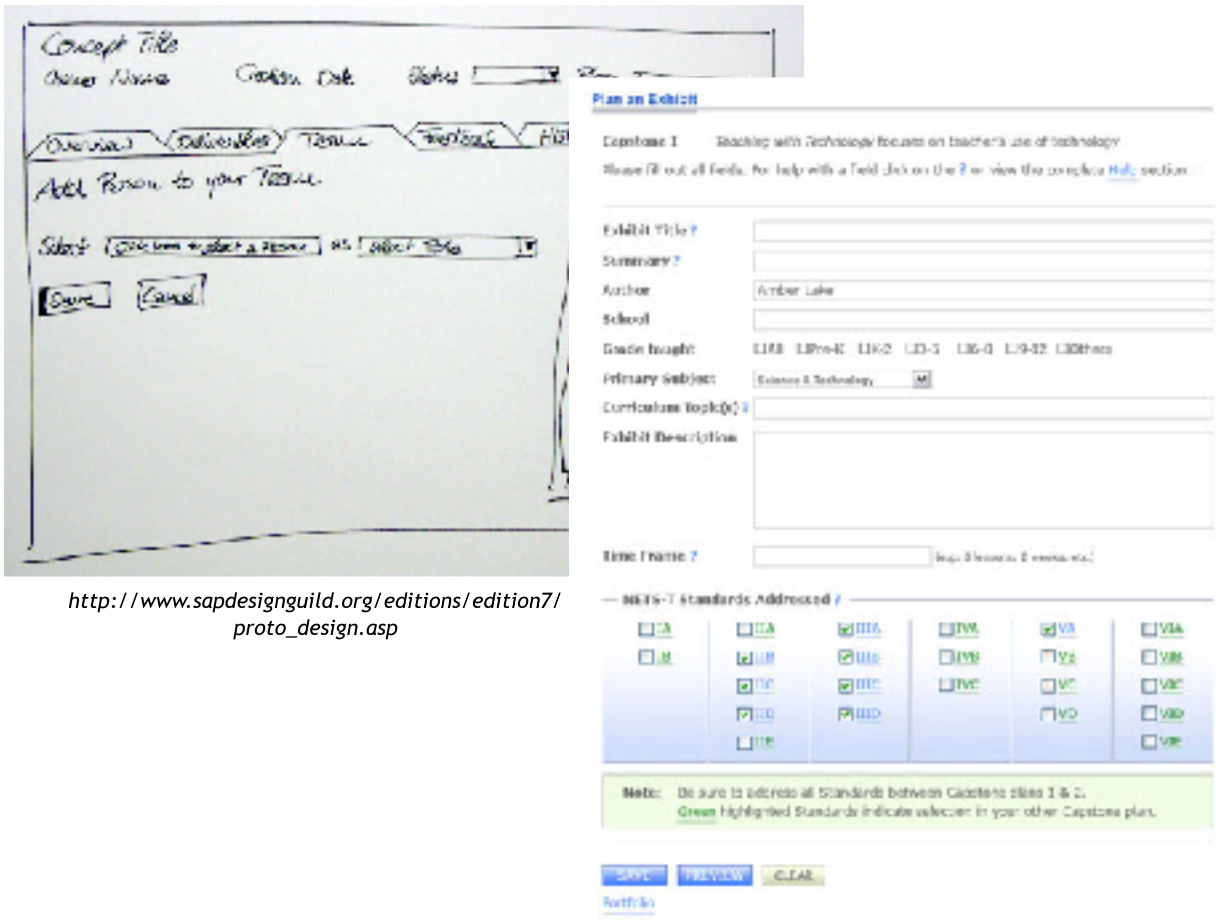
Traditionally, prototypes are categorized as either lo-fi (low fidelity) or hi-fi (high fidelity)

Fidelity can be thought of as how close of an approximation of the final product is being attempted

In this class, we will often use lo-fi as a synonym for paper prototyping and hi-fi as a synonym for digital prototyping

Your design process should use *multiple* levels of fidelity as you move from an idea to a fully designed product

Dimensions of Fidelity



http://www.sapdesignguild.org/editions/edition7/proto_design.asp

Fidelity can be broken down into four basic dimensions:

- Breadth
- Depth
- Look
- Interaction

Breadth

The “breadth” of a prototype refers to how much of the product’s functionality is represented in the prototype

- A very narrow prototype only represents a single feature
- A broad prototype represents all intended functionality

- Prototypes should generally be as broad as needed to cover basic or most important tasks, but not much more

Depth

The “depth” of a prototype refers to how much of the prototype is functional, and how robust it is

- A very shallow prototype has no backend at all and is hard-coded to respond as though the user had provided ideal input
- A deep prototype has some logic and error-handling capabilities
- At first glance, depth may seem unimportant, but it affects the amount of **exploration** a user can do
- Thus depth can actually have a *profound* influence on user testing!

Look

"Look" is probably what most people think of when they think of prototype fidelity

It refers to how accurately a prototype represents the product's intended appearance, including fonts, colors, and graphics

It's generally a good idea to hold off on something that has a high fidelity look until later in the design process

- People are less likely to point out flaws and mistakes
- People can easily fixate on the "little" things
- You are less likely to throw it out and start again

Interaction

"Interaction" refers to how the prototype handles input and output

Interaction can often be simulated

For example, you might create a digital prototype for an iPad application which runs on your desktop and responds to traditional a traditional mouse and keyboard

You might use hyperlinks or animation to simulate clicking interaction (e.g., in Powerpoint)

Hi-Fi Prototyping

Once you have developed a lo-fi prototype and solicited feedback on it through peer critique and user testing:

- You may wish to create another lo-fi prototype
  - (isn’t iterative design fun?)
- Or you may be ready to move on to a hi-fi prototype

Remember, a high fidelity prototype is a substantial time investment!

It is good for evaluating a working design that has been derived from a few rounds paper prototyping

Things you will need to consider…

Choice of tool (more in a minute)

Typography / font

Color palette

Device

Interaction

Implementation

Start by creating a digital version of your paper prototype

Then iterate through user testing and feedback

Don’t design a beautiful prototype that can’t be implemented!

Digital Prototyping Tools

There are literally hundreds…

And more released every day.

I don’t know them all!

What you choose will depend on a variety of factors…

Choosing a tool: Considerations

- Learning Curve
  - How long will it take me to learn this tool?
- Usage
  - Which device will it be used on?
- Fidelity
  - Will it showcase layout structure or complex interactions?
- Sharing
  - Can I collaborate with others on the prototype?
- Cost
  - How much am I prepared to pay for this tool?

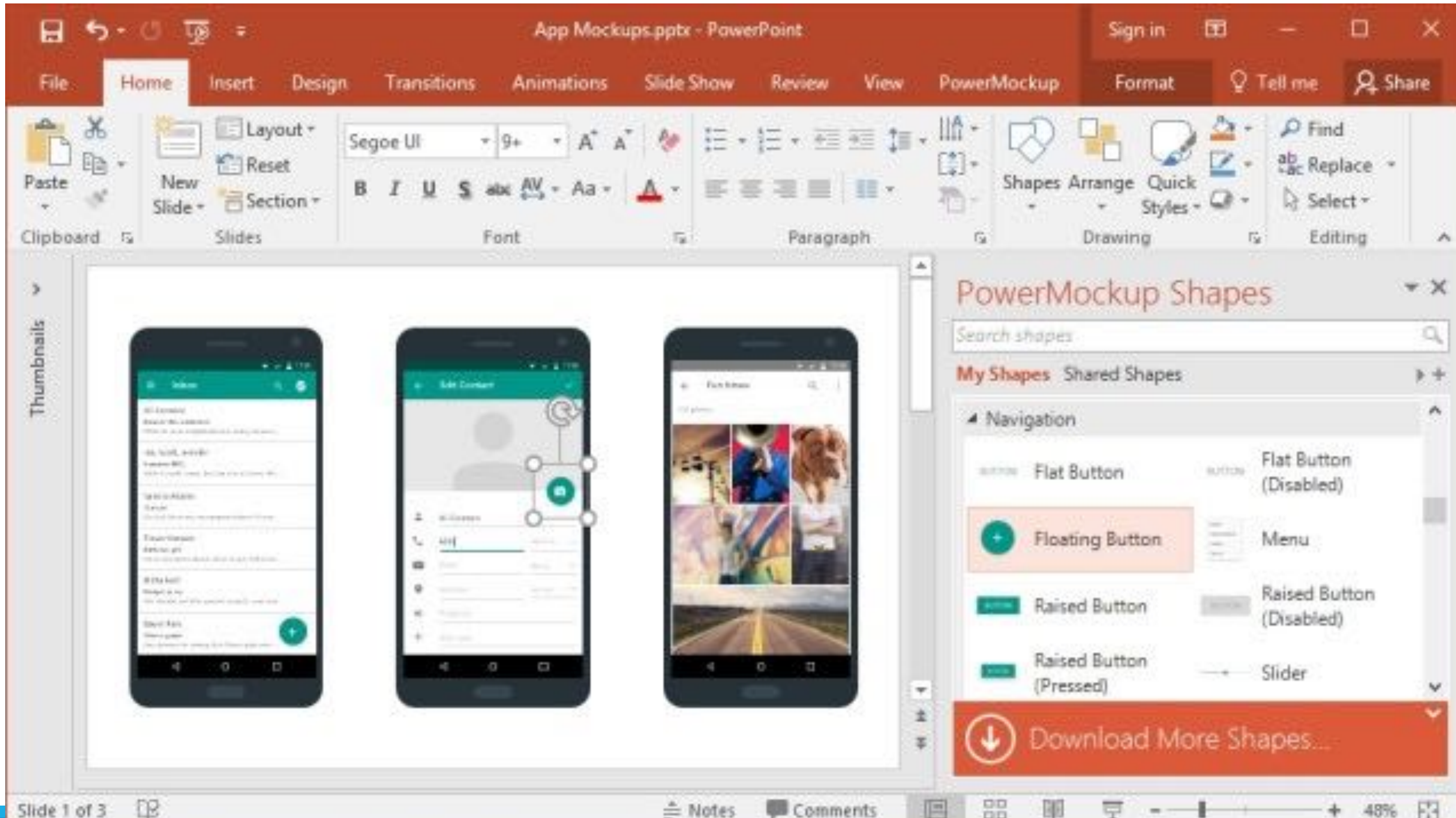
Comparing prototyping tools

A cool tool to help you do this: http://www.prototypr.io/prototyping-tools/

Powerpoint or Keynote

All platforms

Costs money … but many of you have it already

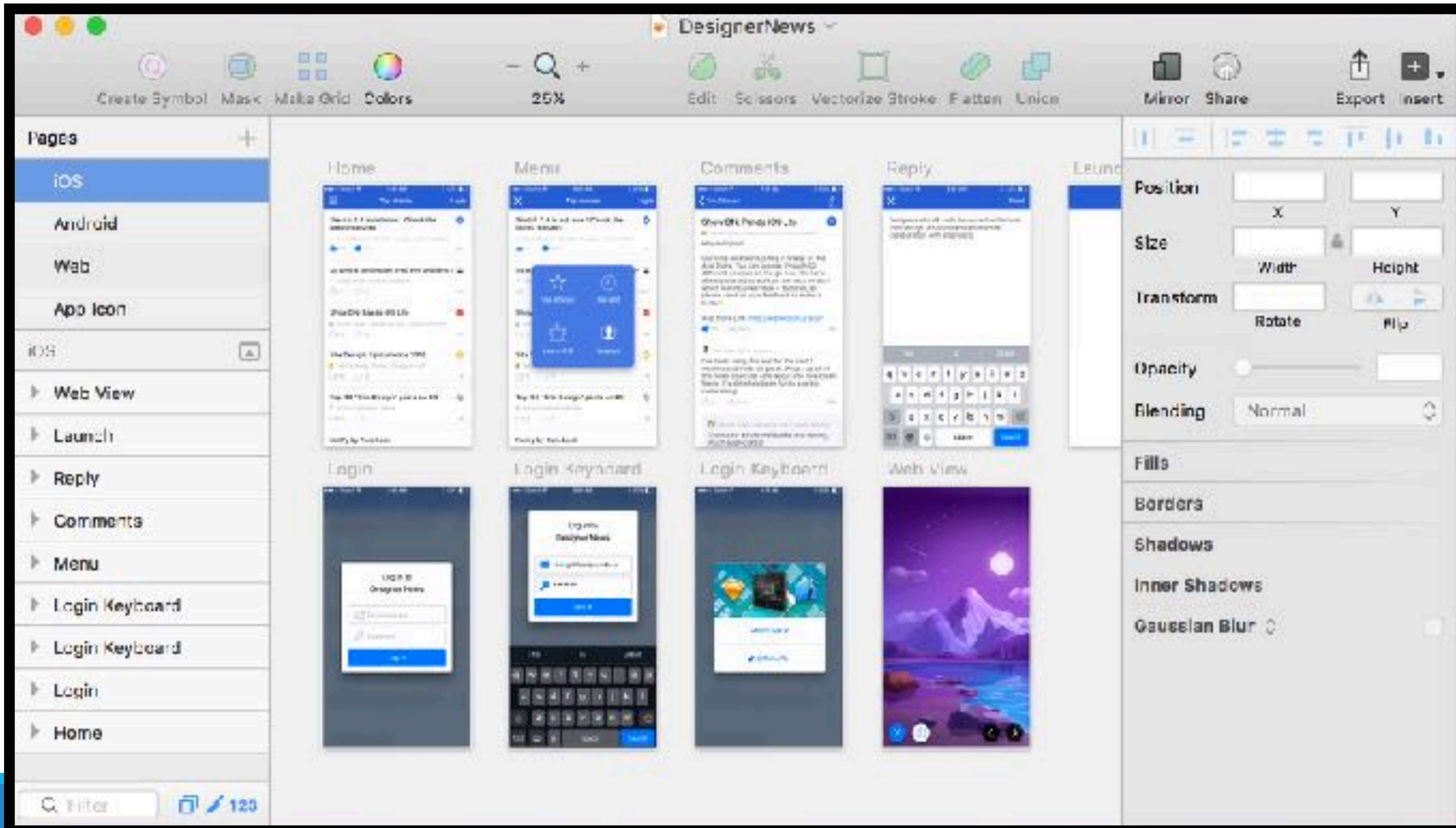


Sketch

Only for Mac

Costs money ($99)

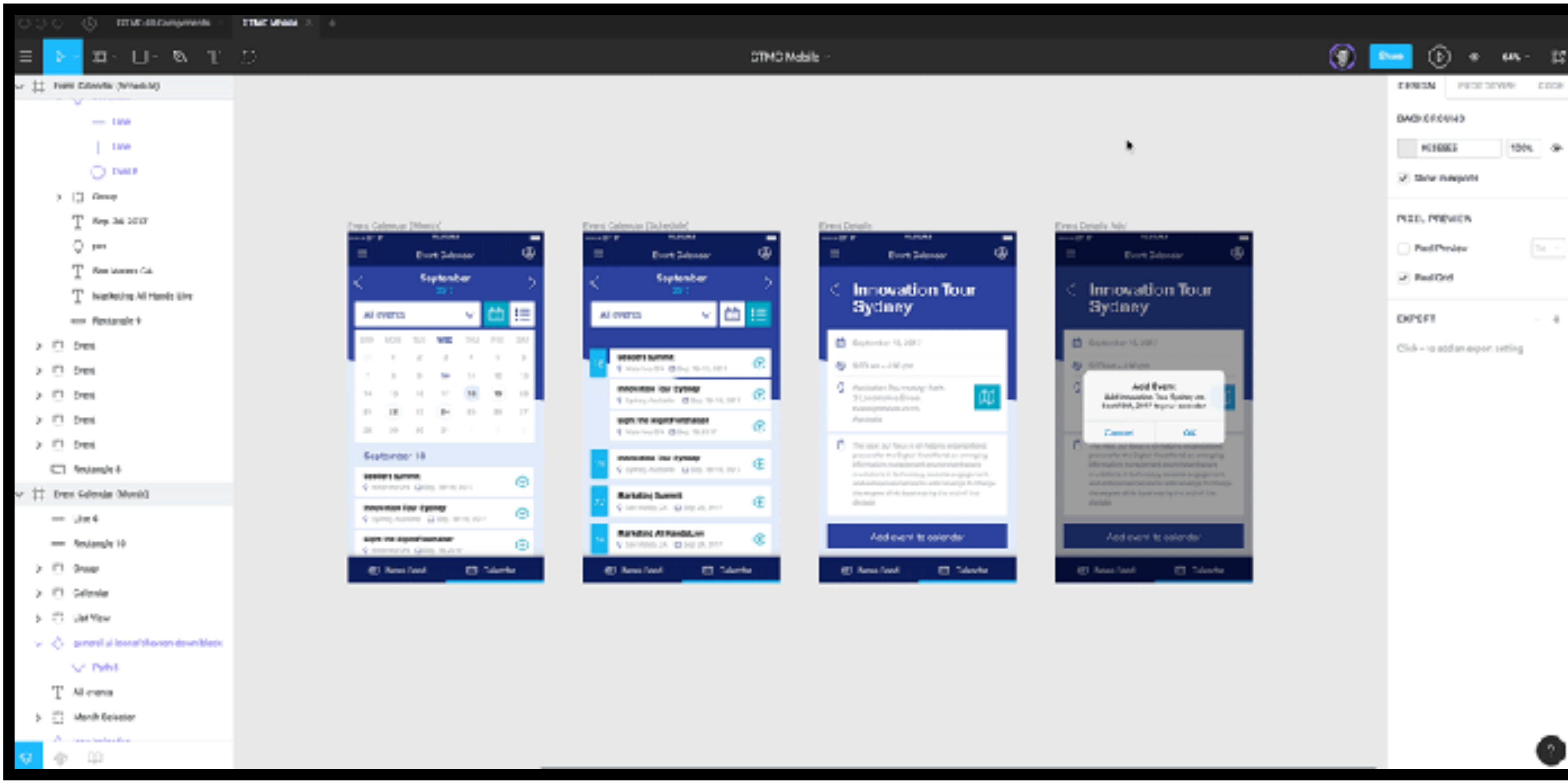
Student discount ($49?)



All platforms

3 projects free

Figma

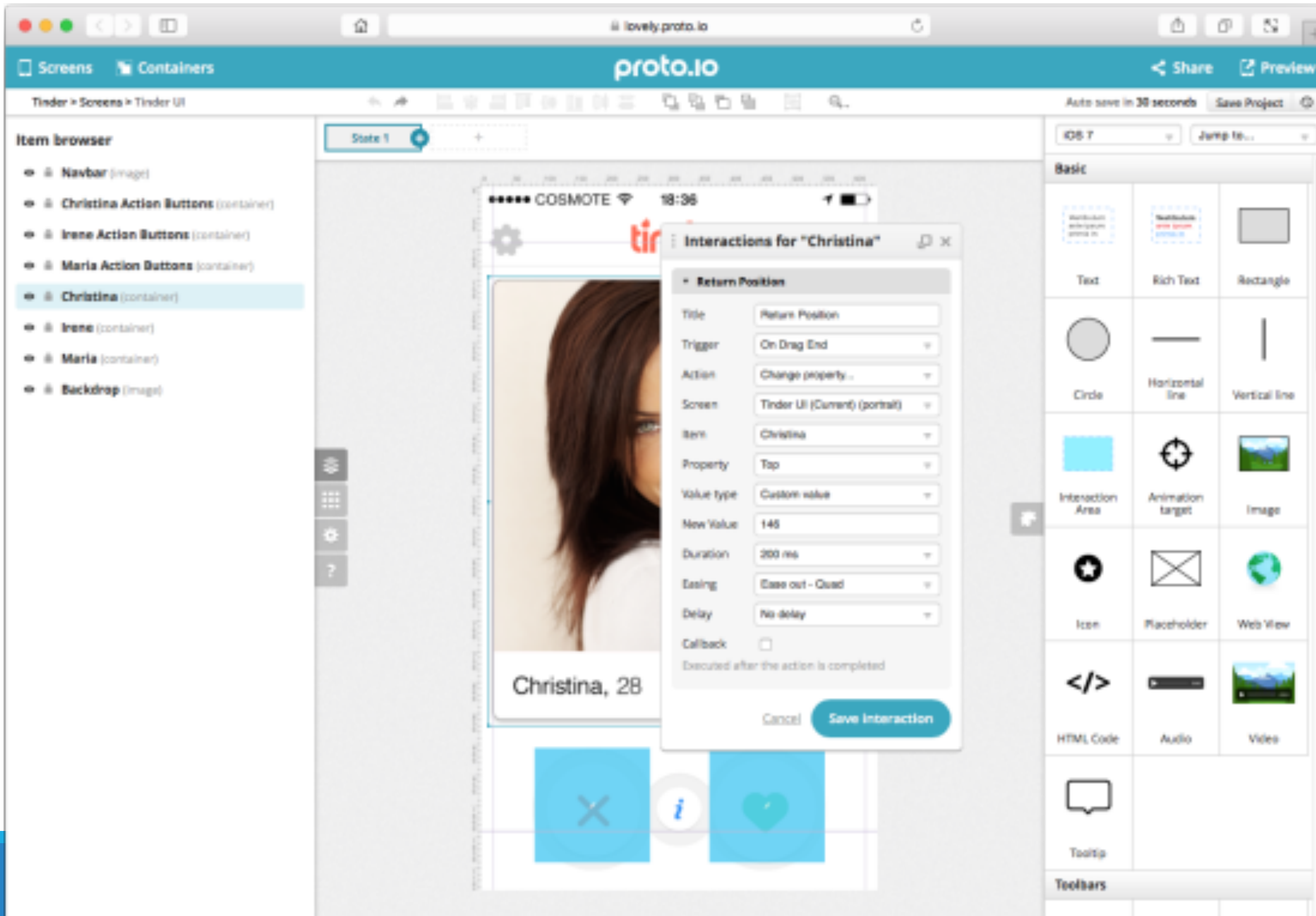


Proto.io

All platforms

Free trial (15 days)

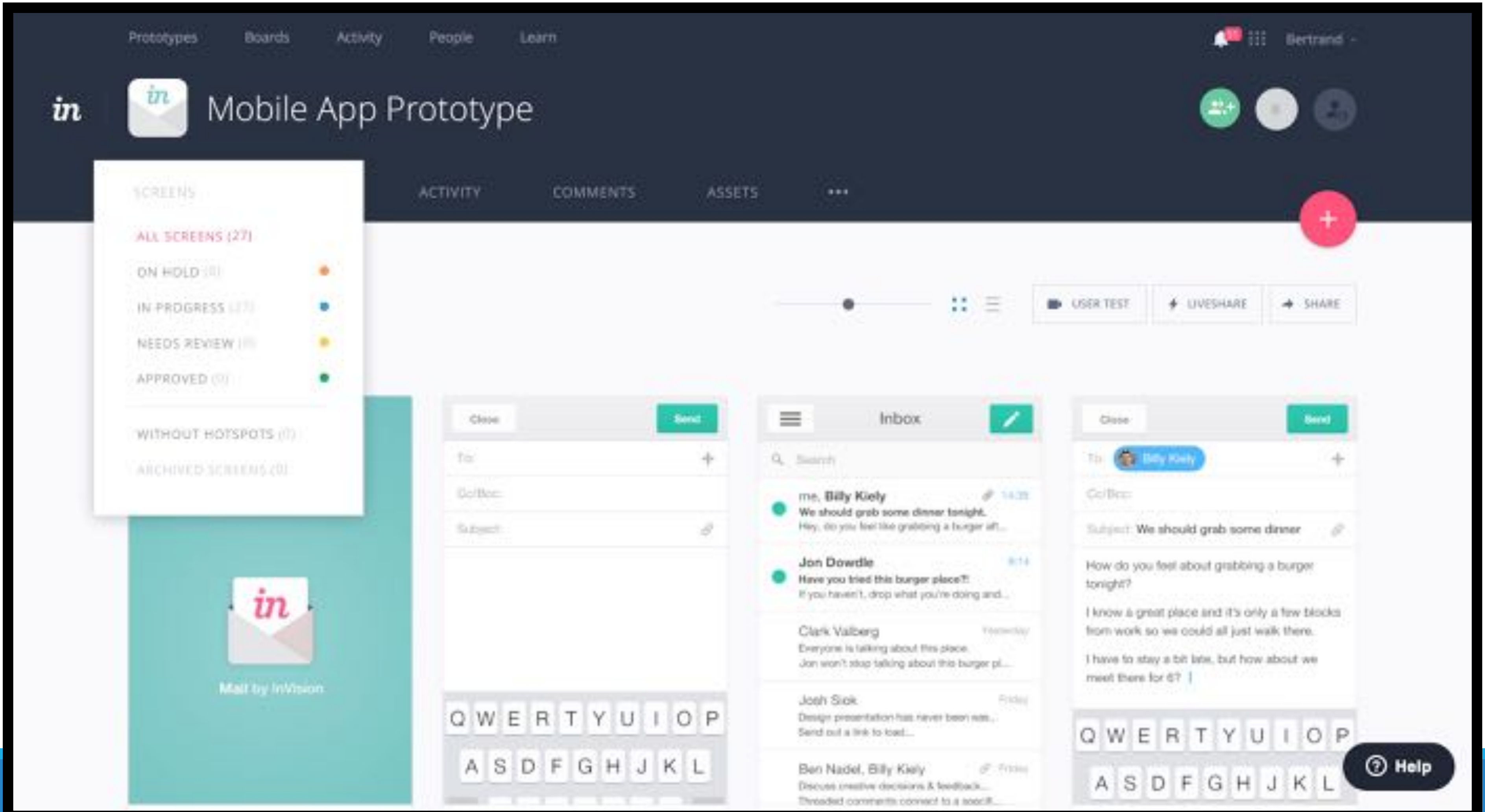
Then monthly fee



InVision

All platforms

One project free, then monthly subscription



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